

# You Can't ABX a Memory

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*New Order, "Bizarre Love Triangle," Glastonbury 1987 vs 2016*

## Abstract

I should say this first, because it changes how you read the rest: I'm not an audio engineer, and I'd bet most people who love this song aren't either. What I have is the song — years of it — and a willingness to measure, honestly, the few simple things that can actually be measured. The thesis here isn't nostalgia. It's that electronic musicians, like any worker, adapt to the tools of their day: the gear changed across thirty years, the rooms changed, the audiences changed, and the band itself changed. So I take one well-documented song — New Order's *Bizarre Love Triangle* (BLT), recorded in 1986 and performed live at the same festival twenty-nine years apart (Glastonbury, 19 June 1987 vs 25 June 2016) — and ask a narrow, physical question: does any of that change show up in the waveform? Each recording is measured *on its own* (loudness and crest factor, brightness, and the steadiness of its own pulse); I never lay one performance over the other. The 1987 audio comes from the edited BBC *Live in Concert* disc; the 2016 from broadcast. Measured that way, the two do differ — the 1987 transfer is louder and flatter, the 2016 brighter and more dynamic, and 1987's own pulse wanders about twice as much — but none of it points cleanly at the medium: both nights ran sequencers, and the dynamics look like the work of the broadcast chain, not the tape. What the numbers can't reach is the feeling. You cannot run a blind test on a memory.

## 1 Introduction: a man asks for less smoke

Seconds before "Bizarre Love Triangle" at Glastonbury in 1987, Bernard Sumner leans into the mic and, deadpan, asks his lighting director — Andy Liddle — to ease off the smoke machine, on the impeccable logic that the crowd has surely *seen* a smoke machine before. It's a small, funny, almost shy moment, and it's the best thirty seconds of stage-talk in the whole night. It is also where my problem starts, because people who have watched the audience video over and over can't agree on what he actually says: some hear "smoke," some hear "small." A roomful of devotees, transcribing the same few mumbled seconds, and they disagree.

That little failure is, in miniature, the whole argument of this essay. If we can't agree on words a man said out loud, into a microphone, right before the song — how sure should any of us be about what the song *sounds* like, twenty-nine years later?

There's an irony sitting over all of it. The band is called *New Order*; the album is called *Brotherhood*. And *Brotherhood* (1986) is the record where the band nearly came apart, so split between its guitar songs and its electronic, sequenced songs that they literally put the rock on one side and the machines on the other. In their own "The Recording of Brotherhood" video (New Order's official channel, 4 October 2024), Bernard describes the stylistic divide — "Hooky didn't like the fact that these sequences were capable of taking over the world" — and how,

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half-joking, they decided to be “completely democratic” with “a rocky side and a sequenced side.” For Peter Hook, writing on machines was a lonelier thing; it broke the way the band had always written together. And yet, at the end of that same video, Hooky says the line that stays with me: when he came to play those songs, he loved them — “more than anything, even more than the acoustic side.” A band named for order, an album named for brotherhood, made by people barely speaking. The divide was real. So was the love.

“Bizarre Love Triangle” turns **forty in 2026**. It was, by the music industry’s own metric, a flop — it failed to chart in the US on its 1986 release and reached only No. 56 in the UK. And yet it became a global, generational anthem: covered by everyone from Frente! to The Killers, sung by people who weren’t born when it came out. Its value never lived in the chart, or in the format.

That last point is really the thesis. New Order belongs to the 1980s synth-pop wave — alongside OMD, Yazoo, Erasure and Pet Shop Boys, with Kraftwerk upstream of all of them — and the defining fact of that music is that it is made on machines that date. Rock hides this: an electric guitar today is essentially the guitar of fifty years ago, and the *vintage* ones cost more, not less. Electronic music can’t hide it. Its instruments are superseded the moment the next box ships, and the musician, like any worker, adapts to the tools of the day. These bands evolved *with* the gear; and because the live rig was largely the studio rig, the era’s recordings sit close to the albums, remixes notwithstanding. So the question isn’t whether the old version had more soul. The question is narrower and physical: the gear changed, the room changed, the audience changed — can we see any of that in the waves?

## 2 The warm sound was already digital

Here’s a fact that complicates the “warm analog” picture before we measure a thing. The shimmer at the front of “Bizarre Love Triangle” was built, in 1986, on a Fairlight CMI — a *digital* sampler, and at the time the most expensive, most prestigious keyboard you could own. Billboard’s appreciation of the song says it outright; so does Sumner, who in a 2005 *Keyboard* interview remembers that they “hired a Fairlight.”

Hook backs up the *prestige* of that machine, if not its specific use on this song. In *Substance* (in the section he calls “Movement Track by Track”), he describes how Martin Hannett wanted Factory’s money spent on a Fairlight CMI Series II — “the same model used by Kate Bush, Jean Michel Jarre and . . . Trevor Horn.” And around the time of the 1987 show — the “True Faith” sessions — the band’s rig was, by Hook’s own list (in the section he titles “Geek Alert: Recording ‘True Faith’”), a wall of digital and hybrid gear: a Yamaha QX1 sequencer, a DX5, an Octave Voyetra 8, an Akai S900 sampler, an Emu Emulator II, an Emu SP12. (That’s the *True Faith* sessions rig, to be clear — not a parts list for the BLT recording.)

I’m not scoring a point. I’m just noting that the “warm analog original” some of us want to defend was, from the start, partly a digital, sequenced record. The band knew it; they argued about it. None of that makes the feeling wrong. It means the warmth, whatever it is, didn’t come from the medium being analog — because it largely wasn’t.

## 3 A note on the voice (1985–1987)

A bit of chronology, because it’s easy to get wrong. Around 1985, during the recording of “Sub-culture,” the producer John Robie told Bernard that the song wasn’t in his key. It landed: it made Bernard realize there was a *right way* to sing — a key the song wanted — and, in a sense, to bend the band toward his own voice. “Bizarre Love Triangle” is 1986. Glastonbury is 1987.

So by the '87 show, Bernard has found his key — his singing has settled in the direction Robie pushed him toward, even if some moments are still openly out of tune, especially in “True Faith,” which they introduce as a first live airing. But settled is not the same as tame, and this is where I want to be careful, because it is a large part of why the night matters so much to people. What the audience video shows is *raw*, and gloriously so. Bernard is loose and plainly enjoying himself — the comments are sure he’s on something, and whatever it is, he’s joking, throwing whoahs, singing ahead of the beat and somehow making it work, and screaming — not just here but right across the set, the kind of thing the comments keep coming back to. It’s synth-pop handed a jolt of post-punk energy. The people under that video aren’t wrong to love it for exactly that — “uncut, raw, timeless,” one of them calls it. The rawness isn’t a flaw in the performance; for a lot of fans it *is* the performance. The Robie thread, and Hook’s discomfort with the machines, is mid-1980s context — how they got *to* 1987. It is not a verdict on how 1987 feels.

## 4 Glastonbury, and how this night even survives

It’s worth remembering how thin the record is for a night like this. Glastonbury began in 1970, on Michael Eavis’s dairy farm, the weekend after Jimi Hendrix died, and for years it was a small thing. It swelled through the 1980s and became the giant we know in the 1990s — and sustained television coverage is itself a 1990s arrival. Which is exactly why a 1987 set like this one survives mostly through amateur recordings, with video rarer still.

This night did get an official release, of a sort: the BBC’s *Radio 1 Live in Concert* (Windsong International, 1992), an edited nine-track disc. It keeps “Bizarre Love Triangle” (track six) and “True Faith,” but it drops four songs from the live set — “Elegia,” “Ceremony,” “Sub-culture” and “Sunrise.” The complete concert survives at all only because someone synched an audience video to the BBC soundboard and put it online: the YouTube channel Soundsville International (1987 footage). Without that upload, there would be no visual record of the night.

A method note, then: I use that video only for what’s *said* on stage — the talk between songs. For anything about the actual *sound*, I use the edited CDs.

## 5 What I actually wanted to measure

Forget the engineering vocabulary for a second. Here’s the song as a body feels it.

The drum that kicks in after that opening roll, hitting harder and more present than you expect — the more it’s there, the more it booms. The synth figure that runs through the whole thing, that *turun-turun-turn* anyone who knows the song is already hearing — so present in the eighties versions. Hook’s bass, doing what Hook’s bass does, carrying the actual melody instead of just holding the bottom. The arpeggio that beats under all of it like a heart and pulls the song forward. And the ending, where the bass and Bernard’s guitar take it home.

That’s the heart of the song. My only real question is small and stubborn: **does any of that show up in the waveform?** If you strip away the engineering talk and just look at the shapes the air made — is the punch of that drum, the pulse of that arpeggio, the line Hook’s bass draws, actually *in* there? Or is it something we add ourselves, on the other side of the speaker? I genuinely don’t know, so I wrote the analysis to look for exactly those things — per recording.

## 6 What can honestly be measured — and how

A few basic things, in plain words, with the technical names kept where they earn their place:

- **How loud, and how punchy-vs-flat** — integrated loudness, and the *crest factor* (the gap between the loudest spikes and the average). A drum that really hits leaves big spikes; a heavily squashed “brickwall” master flattens them.
- **How bright** — the *spectral centroid* (where the energy’s centre of mass sits) and the spectral tilt (how fast it rolls off). Up means bright, down means darker.
- **How steady its own pulse is** — within *one* recording, beat to beat: does its tempo hold or wander.

To make “punchy vs flat” concrete, I ran it on a *fake* signal I built from scratch (no real music): mastered gently, the loud-vs-average gap is about 19 dB; squashed with a modern brickwall master it collapses to about 4 dB. Same for drift: wobble a synthetic tape and it wanders about 7.4 ms; a digital clock holds at 0.000. **Those are toy demonstrations of the mechanism on a synthetic signal — not measurements of “Bizarre Love Triangle.”** They just make the words mean something before we look at the real thing.

### The real song

Here is what the two recordings actually measure, each on its own (Table 1; the per-recording traces are in Figures 1 and 2). I’ll read it plainly. The 1987 BBC transfer is the louder and flatter of the two — about  $-12.5$  LUFS, with a mean crest factor near 11.3 dB — while the 2016 broadcast is quieter and keeps its transients more open (about  $-17.8$  LUFS, crest near 13.9 dB). On brightness it runs the opposite way to what “old and warm” might suggest: the 2016 capture sits higher, a spectral centroid around 842 Hz against 608 Hz in 1987. A caution worth keeping: these are two different captures of different lengths — the 1987 disc track and the longer 2016 broadcast — so the numbers describe *these recordings*, not the medium. The flatter, louder 1987 is the fingerprint of heavier limiting somewhere in the chain — plausibly the 1980s FM broadcast path — not of tape versus transistor.

Table 1: Each recording measured on its own. Nothing is implied beyond reading two independent characterisations.

| Measure                              | 1987 (BBC disc) | 2016 (HD broadcast) |
|--------------------------------------|-----------------|---------------------|
| Integrated loudness (LUFS)           | $-12.5$         | $-17.8$             |
| Mean crest factor (dB)               | 11.3            | 13.9                |
| Spectral centroid, $<15$ kHz (Hz)    | 608             | 842                 |
| Spectral tilt (dB/octave)            | $-5.4$          | $-6.4$              |
| Own pulse — tempo (BPM)              | 117             | 123                 |
| Own pulse — inter-beat $\sigma$ (ms) | 20              | 10                  |

Two small things hold steady across both nights, and I like that they do: the recurring synth figure comes out at essentially the same rate, and the ending stays bass-led — the low end sitting a few decibels above the guitars on both — which is the shape Hook’s bass gives the song.

## Timing, on its own terms

The pulse is measured *inside* each recording, never one against the other. In 1987 the song holds a tempo near 117 BPM with a beat-to-beat spread of about 20 ms; in 2016 it sits near 123 BPM with about 10 ms. So 1987’s own pulse wanders roughly twice as much as 2016’s. I’m going to leave that as a number, not a story. I won’t call one “breathing” and the other “rigid”; I won’t pin it on analog-versus-digital — both nights ran sequencers, the machines were on stage in 1987 too; and I’ll keep in mind that, in the lab, *adding* timing wander made listeners feel *less* groove, not more (Davies and colleagues, 2013). What the wander is, and whether anyone feels it, is not something this measurement can reach.

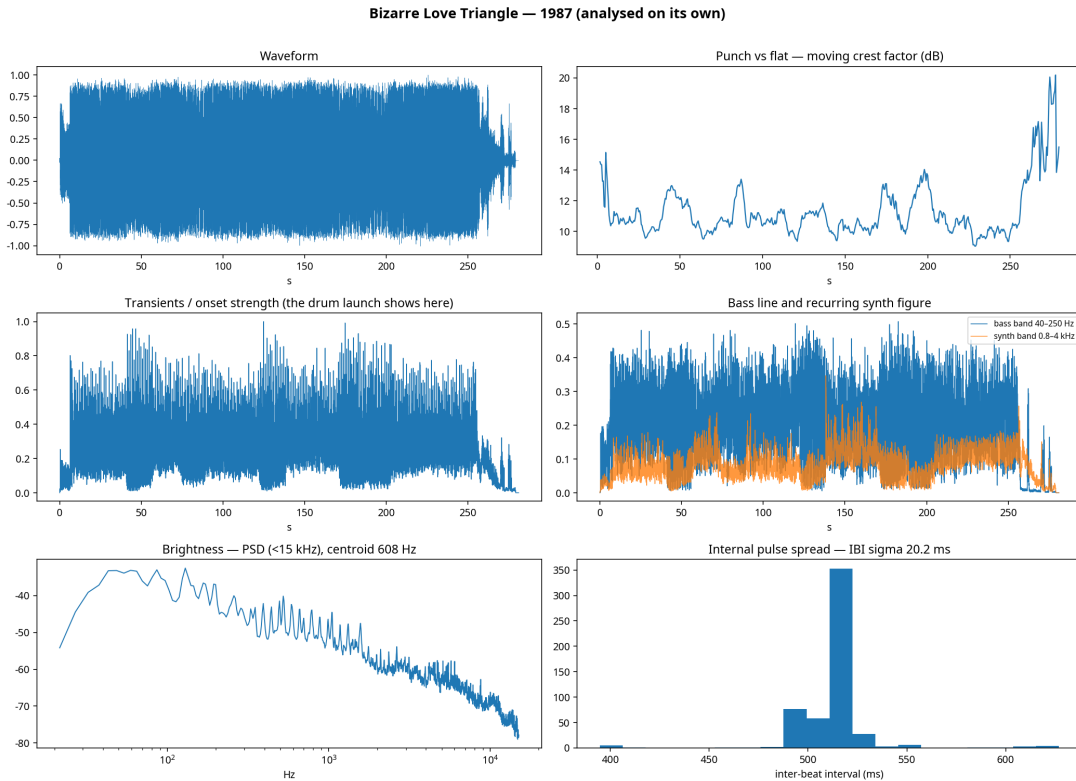


Figure 1: The 1987 recording, measured on its own.

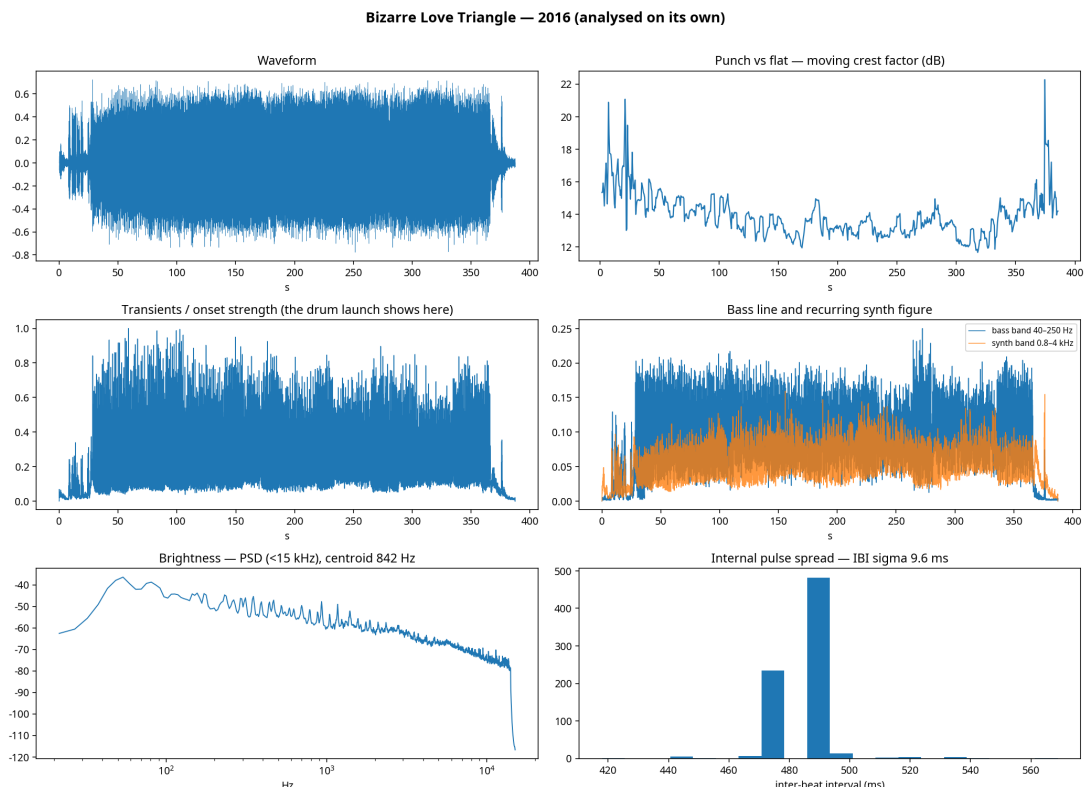


Figure 2: The 2016 recording, measured on its own.

## 7 The two nights

By 2016 almost everything has changed except the song. Hooky is gone — he left in 2007 — so the bass line that drew the melody is someone else’s now. It’s an HD BBC capture, a modern line-array PA, a vast modern crowd. And Bernard no longer asks anyone to turn the smoke down; he asks the crowd to *move* — “come on, this is Glastonbury.” Same song, two technological eras, two audiences, and a band that is not quite the same band.

The recordings do sound different, and the measurements agree — but the differences sit in the master and the broadcast chain (the loud, flat 1987 transfer against the more open 2016 one), in the gear and the room, in the mix, and in the band itself. They do not sit in “analog versus digital”: both nights were sequenced. The tools and the audience changed across thirty years, and some of that change is visible in the waves. What it *means* — whether any of it is the thing we actually feel — the numbers can’t say. That part stays in memory.

## 8 Conclusion

The belief that analog is warm and digital is cold doesn’t quite come through this one song unchanged — but that’s a remark about one song, not a verdict on the feeling, and not a verdict on you for having it. The warmth was a digital sampler. The differences I can measure — louder and flatter then, brighter and more open now, a pulse that wanders more in 1987 — line up with the master and the broadcast chain and the room, not with analog versus digital.

I set out to see whether the heart of the song — the drum, the arpeggio, Hook’s bass, the way it ends — is visible in the waveform, or whether we put it there ourselves. I don’t have that answer yet. What I do know is that you can measure the air, but you cannot measure being

nineteen. You cannot run a blind test on a memory. I'd rather end on that honest gap than on a verdict I can't support.

## References

*Audio used for sound analysis: the edited CDs. Video used only for on-stage talk.*

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